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I am a third year student pursuing an offline degree(GTU-BE in CS) with iit BS degree, Currently exploring areas of interest in traditional development as well as ML.

AI/LLM usage

Have used AI for PDF generation syntax(fpdf), initial axios/router setup and also in the pdf loading part. All include shall not exceed 5-10%.

Description

Providing a platform for smooth job posting and application with the goal of ensuring smooth placement of students. Admin shall ensure management of students, companies and job drives.

Technologies used

Flask -> Backend framework

Flask jwt -> Stateless authorization and authentication

Flask Mail -> Smtplib for sending mails

Flask Redis -> Queue for async jobs + Cache Management

Celery -> Async Jobs, workers + scheduled jobs

Sqlalchemy -> DB creation and query management

Fpdf2 -> Pdf creation

Vue -> Frontend framework, router for route management

Axios -> API management

HTML -> Templates

Bootstrap + CSS -> styling

DB Schema Design

User table

Column	Data type	Constraint
id	Integer	Primary Key
name	String(30)	Not null
email	String(30)	Unique, Not null
password_hash	String(256)	Not null
role	Enum(UserRole)	Default='student', Not null
created_at	DateTime	Default=current time

Student table

Column	Data type	Constraint
id	Integer	Primary Key
user_id	Integer	Foreign Key, Unique, Not null
cgpa	Float	Not null
branch	String(20)	Not null
graduation_year	Integer	Not null
skills	String(300)	
resume_path	String(100)	
is_blacklisted	Boolean	Default=False, Not null

Company Table

Column	Data type	Constraint
id	Integer	Primary Key
user_id	Integer	Foreign Key, Unique, Not null
is_approved	Enum(ApprovalStatus)	Default='pending', Not null
hr_email	String(30)	Not null
description	Text	Not null
industry	String(50)	Not null
location	String(50)	
website_link	String(30)	
is_blacklisted	Boolean	Default=False, Not null

Job Position Table

Column	Data type	Constraint
id	Integer	Primary Key
company_id	Integer	Foreign Key, Not null
title	String(100)	Not null
description	Text	Not null
deadline	DateTime	Not null
skills_required	String(300)	Not null
vacancies	Integer	Default=1, Not null
status	Enum(JobStatus)	Default='pending', Not null
min_cgpa	Float	Not null
eligible_branches	String(200)	Not null
eligible_graduation_years	String(100)	Not null

Column	Data type	Constraint
created_at	DateTime	Default=current time

Application Table

Column	Data type	Constraint
id	Integer	Primary Key
student_id	Integer	Foreign Key, Not null
job_id	Integer	Foreign Key, Not null
applied_at	DateTime	Default=current time
status	Enum(ApplicationStatus)	Default='applied', Not null
interview_date	DateTime	
interview_location	String(100)	
salary	Integer	
joining_date	Date	
feedback	Text	
offer_letter_path	String(100)	

Placement Table

Column	Data type	Constraint
id	Integer	Primary Key
student_id	Integer	Foreign Key, Unique, Not null
company_id	Integer	Foreign Key, Not null
job_id	Integer	Foreign Key, Not null
job_title	String(100)	Not null
salary	Integer	Not null
joining_date	Date	Not null
placed_at	DateTime	Default=current time
placement_letter_path	String(100)	

Relationships between tables:

- User-Student: 1-1
- User-Company: 1-1
- Company-JobPosition: 1-N
- JobPosition-Application: 1-N
- Student-Application: 1-N (but with constraint, a student can have single application for a job)
- Student-Placement: 1-1
- Company-Placement: 1-N
- JobPosition-Placement: 1-N



API Design

Yaml has not been created(optional)

Auth

Method	Endpoint	Description
POST	/api/auth/register/student	Register student
POST	/api/auth/register/company	Register company
POST	/api/auth/login	Login (all roles)

Admin(/api/admin)

Method	Endpoint	Description
GET	/admin/stats	Platform statistics
GET	/admin/job/<job_id>/stats	Job statistics

Method	Endpoint	Description
GET	/admin/companies	All companies
GET / PATCH	/admin/company/	Get / approve / reject company
PATCH	/admin/company/blacklist-status	Toggle blacklist
GET	/admin/jobs	All jobs
GET / PATCH	/admin/job/	Get / approve / reject job
GET	/admin/students	All students
GET / PATCH	/admin/student/	Get / blacklist student
GET	/admin/student/resume	Download resume
GET	/admin/student/applications	Student applications
GET	/admin/applications	All applications
GET	/admin/application/	Specific application
GET	/admin/application/offer-letter	Download offer letter
GET	/admin/placements	All placements
GET	/admin/placement/letter	Download placement letter
POST	/admin/trigger/interview-reminder	Trigger reminder task
GET	/admin/task/<task_id>	Task status

Company(/api/company)

Method	Endpoint	Description
GET / PATCH	/company/profile	Get / update profile
GET	/company/stats	Company statistics
POST	/company/job	Create job posting
GET / PATCH	/company/job/	Get / open / close job
GET	/company/applications	All applications
GET	/company/job/applications	Applications for a job
GET / PATCH	/company/application/	Get / update status
PATCH	/company/application/interview	Schedule interview
GET	/company/application/offer-letter	Download offer letter
GET	/company/resume/<student_id>	Download student resume
GET	/company/placements	All placements
GET	/company/placement/	Specific placement
GET	/company/placement/<student_id>/letter	Download placement letter
POST	/company/report	Trigger monthly report
POST	/company/export/csv	Export data to CSV

Student(/api/student)

Method	Endpoint	Description
GET / PATCH	/student/profile	Get / update profile + resume
GET	/student/jobs	All open jobs
GET	/student/job/	Specific job
POST	/student/job/apply	Apply for a job
GET	/student/applications	All applications
GET / PATCH	/student/application/	Get / accept / reject offer
GET	/student/application/offer-letter	Download offer letter
GET	/student/placement-letter	Download placement letter
GET	/student/resume	Download own resume
POST	/student/export/csv	Export data to CSV

Architecture and Features

Visual representation of project directory

PlacementPortalApplicationV2/

```
/backend
  /routes
  /services
  /instance
  /static
    /pdfs
    /reports
    /resume
/frontend
  /src
    /views
    /router
    /services
```

Note: A lot more files are there such as .gitignore, README.md, etc that I have not represented here.

Main feature: Interview reminders, application history and monthly reports shared through emails with smtp.

Video

Link: <https://drive.google.com/file/d/1cgApGbHEVSQaomlYwmEYPxZc4MJkhGvq/view?usp=sharing>